

SOP-PHARMA-009 – AI-Based Batch Monitoring

Version: 2.0 | Status: Effective | Department: Quality Assurance (QA) / Production / IT

1. Purpose

This SOP defines the procedures and compliance requirements for the use of AI-based batch monitoring systems in pharmaceutical manufacturing environments.

2. Scope

- AI-based monitoring systems
- Production batch review workflows
- GMP-regulated production lines
- Audit and compliance activities

3. Responsibilities

Role	Responsibility
QA Team	Review and approve batch release decisions
IT Team	Maintain infrastructure and ensure cybersecurity
Production Team	Monitor process deviations and alarm notifications
Validation Team	Validate AI models and workflows

4. Procedure

1. AI models must be validated before deployment into production.
1. Ensure that all production data originates from validated systems.
1. Review AI-generated alerts before approving batch release.
1. Maintain complete audit logs for all AI-driven decisions.
1. Conduct annual reviews of AI model performance and effectiveness.

5. Deviations

Deviation ID	Description	Status	Cause
DEV-AI-001	Missing audit logs	Open	Logging service failure
DEV-AI-002	Invalid model deployment	Closed	Approval workflow bypass
DEV-AI-003	Unauthorized API access	Under Review	API misconfiguration

6. CAPA Actions

- Implement immutable audit log storage.
- Require mandatory QA approval before any model deployment.
- Enable Multi-Factor Authentication (MFA) for all AI system users.

7. Audit Findings

AUD-AI-009 – FDA Inspection Finding: The AI deployment process lacked a documented approval workflow.

8. Decisions

Management approved the implementation of centralized AI governance and mandatory human-in-the-loop review for all critical batch-related decisions.

9. Approval

Field	Value
Prepared By	AI Compliance Team
Reviewed By	QA Manager
Approved By	Head of Quality
Effective Date	20-May-2026
Next Review Date	19-May-2027